

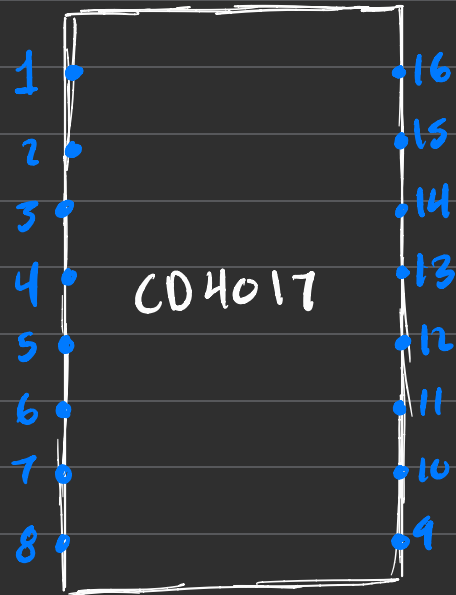
CD4017

- The CD4017 compared to the 555 the CD4017 is much more state-based
- At the highest level, it's entire job is to cycle through outputs

Pulse $\rightarrow$ Q0 $\rightarrow$ Pulse $\rightarrow$ Q1 $\rightarrow$ Pulse $\rightarrow$ Q2...

- It only can have one HIGH output at a time

- | | |
|-------------------|---------------|
| 1. Q5 | 2. Q1 |
| 3. Q0 | 4. Q2 |
| 5. Q6 | 6. Q7 |
| 7. Q3 | 8. GND |
| 9. Q8 | 10. Q4 |
| 11. Q9 | 12. Carry Out |
| 13. Clock Inhibit | 14. Clock |
| 15. Reset | 16. Power |



Clock:

- Arguable the most important pin
- The clock is move of a counter, when clock signal makes a rising transition the counter increases by one

Start: Q0 - HIGH Remember only 1 HIGH
Clock ↑: Q1 - HIGH at a time
Clock ↑: Q2 - HIGH

- It responds to a **RISING EDGE** not just a signal

Clock Inhibit:

- Think of it as a pause button on the clock

Pin 13 LOW → Counting Enabled
Pin 13 HIGH → Counting Inhibited

Reset:

- This reset is **ACTIVE HIGH**
- If LOW the clock counts normally
- When HIGH the counter is set to Q0 on repeat until it is LOW

Q (0-9):

- These are all output pins and are very annoyingly scrambled

Q0 = pin 3

Q1 = pin 2

Q2 = pin 4

Q3 = pin 7

Q4 = pin 10

Q5 = pin 1

Q6 = pin 5

Q7 = pin 6

Q8 = pin 9

Q9 = pin 11

- Only the output the current goes through is set to HIGH, all the other are LOW

Carry out :

- This is an output that is used different then $Q_0 - Q_9$

- It produces a slow repeating signal that trigger every 10 clock ticks

10 counts $\rightarrow$ 1 Carry out

- Typically used in some chain of CD4017

Fast Counter $\rightarrow 0-9 \rightarrow$ slow Counter $+1$